KSR Annual Report - 2022

Korean Stroke Registry

About Korean Stroke Registry (KSR)

Korean Stroke Registry(KSR) is a representative multi-center, prospective, and hospital-based stroke registry in Korea. KSR established to collect data on patients with acute stroke and transient ischemic attack, and to monitor, evaluate, and improve the quality of stroke care in Korea.

In 2001, KSR was launched with 33 major hospitals from 23 medical schools with support from the Korean Stroke Society. In 2006, KSR was progressed with the participation of 66 major hospitals, as one of the six divisions of the Clinical Research Center for Stroke (CRCS) with the support of the Ministry of Health and Welfare, Republic of Korea.

In 2016, the Korean Stroke Society established the KSR steering committee and is operating a new KSR system with the support of the Korea Disease Control and Prevention Agency, Republic of Korea.

As of 2020, 84 hospitals nationwide are participating in the KSR, and 62 hospitals are participating in the KSR Core DB, and participating hospitals are gradually expanding.

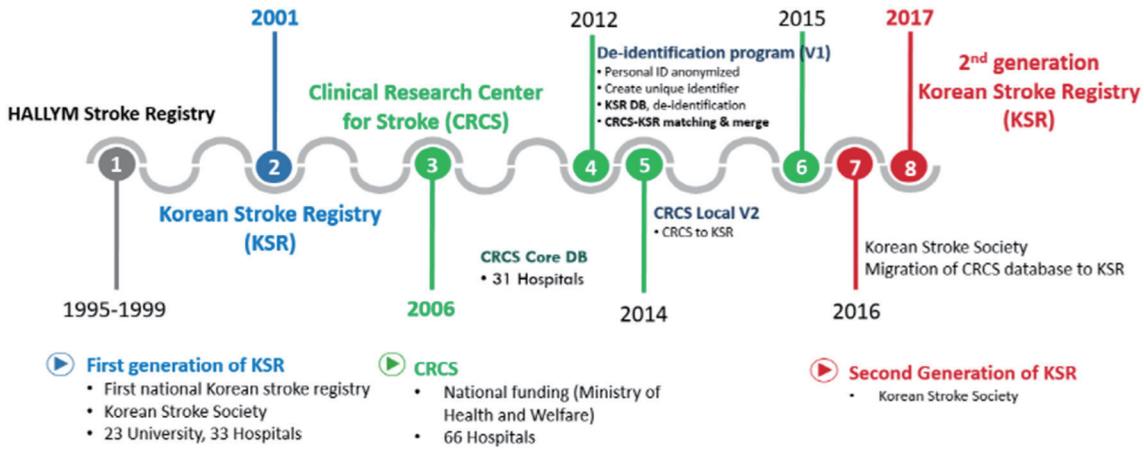

Figure 1. Historical Review of Korean Stroke Registry

Goals of KSR

The overall goal of the KSR is to provide reliable and representative national data on acute stroke care and outcome of patients with all stroke or transient ischemic attack cases in Korea

- To assess the characteristics and secular trend of acute stroke in Korea
- To monitor the quality of acute stroke care
- To implement professional guidelines for acute stroke care
- To monitor the impact of care on post discharge patient outcomes
- To develop a large comprehensive data set for stroke research

Database of KSR

The database of KSR composes a comprehensive and integrated patient management system by supplementing the problems of existing disease-specific and research-specific clinical databases. Reflecting the medical reality, the Core DB was configured to allow input of common data in multiple medical institutions, and the Extended DB was configured to allow data to be added in each study. It is also configured to add long-term prognosis and functional status information (Outcome DB).

In order to manage the data quality of KSR, data are backed up in real-time, and an error check system is automatically applied to reduce errors that may occur during the data entry process. All personal information data has been completely removed through a de-identification program. In addition, the quality of data is improved by managing missing and abnormal input values as a query through the quality management of the database.

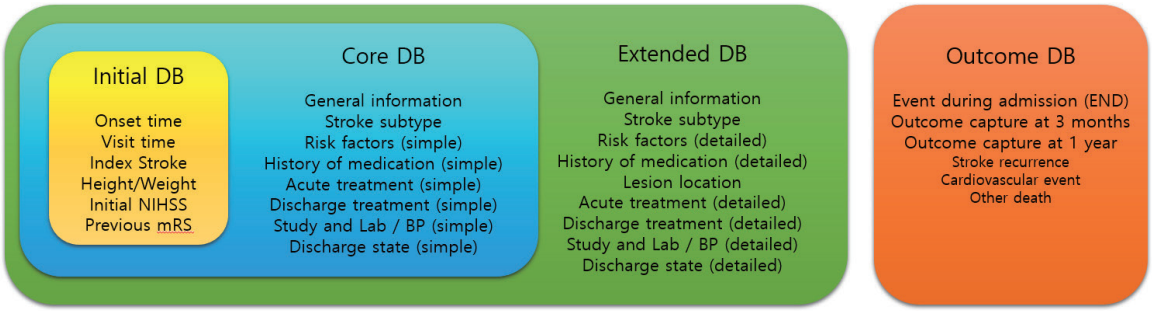

Figure 2. Database structure of Korean Stroke Registry

Participating stroke centers

(KSR Core DB, 62 hospitals, alphabetical order)

The KSR annual report 2022 analyzed the cases of 42 hospitals (marked with an asterisk) that continuously participated in the audit process from Jan 2012 to Dec 2020.

- Asan Medical Center*
- Bucheon Sejong Hospital
- Bundang Jesaeng Hospital*
- Cha University Bundang Medical Center
- Changwon Fatima Hospital*
- Chonnam National University Hospital*
- Chung-Ang University Hospital*
- Chungbuk National University Hospital*
- Chungnam National University Sejong Hospital*
- Daegu Caholic University Medical Center*
- Daejeon Eulji Medical Center, Eulji University*
- Dong-A University Hospital*
- Dongguk University Ilsan Hospital*
- Dongkang Hospital
- Gachon University Gil Medical Center*
- Gangnam Severance Hospital*
- Gyeonggi Provincial Medical Center Icheon Hospital
- Gyeongsang National University Hospital*
- Gyeongsang National Universtiy Changwon Hospital*
- Hallym University Chuncheon Sacred Heart Hospital*

- Hallym University Dongtan Sacred Heart Hospital*
- Hallym University Sacred Heart Hospital*
- Hana General Hospital
- Hanyang University Hospital*
- Incheon Paik Hospital
- Inha University Hospital*
- Inje University Busan Paik Hospital*
- Inje University Ilsan Paik Hospital*
- Jeju National University Hospital*
- Kangbuk Samsung Hospital
- Kangdong Sacred Heart Hospital*
- Kangwon National University Hospital*
- Keimyung University Dongsan Hospital*
- Konkuk University Medical Center*
- Korea University Anam Hospital
- Korea University Ansan Hospital
- Korea University Guro Hospital*
- Kosin University Gospel Hospital*
- Kyung Hee University Medical Center*
- Kyungpook National University Chilgok Hospital*
- Mokpo Jung-Ang Hospital
- Myongji Hospital
- Myongji St. Mary's Hospital
- National Health Insurance Service Ilsan Hospital
- Nowon Eulji Medical Center, Eulji University*
- Pohang Stroke and Spine Hospital*
- Pusan National University Hospital

- Samsung Medical Center*
- Seoul Medical Center*
- Seoul National University Bundang Hospital*
- Seoul National University Hospital*
- Severance Hospital
- SMG-SNU Boramae Medical Center*
- Soonchunhyang University Hospital Bucheon
- Soonchunhyang University Hospital*
- The Catholic University of Korea Seoul St. Mary's Hospital
- The Catholic University of Korea St. Vincent's Hospital
- Ulsan University Hospital*
- Wonju Severance Christian Hospital
- Yeungnam University Medical Center*
- Yonsei S Hospital
- Yuseong Sun Hospital*

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Abbreviations

BMI	Body mass index
DBP	Diastolic blood pressure
HDL	High-density lipoprotein
IQR	Interquartile range
LDL	Low-density lipoprotein
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
SBP	Systolic blood pressure
SD	Standard deviation
TG	Triglyceride
TOAST	Trial of ORG 10172 in Acute Stroke Treatment

Global Statistics of KSR

Demographic profile of acute ischemic stroke

- Of 122,099 stroke cases registered in 42 hospitals between Jan 2012 and Dec 2020, number of acute ischemic stroke was 108,480 (88.8%).
- Men were 63,886 (58.9%) and women were 44,594 (41.1%).
- Mean (SD) age at onset was 65.9 (12.5) for men and 72.1 (12.7) for women.

Table 1. Number of registered acute stroke patients

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Sum
2012	887 (8.7)	841 (8.3)	928 (9.1)	790 (7.8)	857 (8.4)	884 (8.7)	897 (8.8)	800 (7.9)	815 (8.0)	876 (8.6)	833 (8.2)	746 (7.3)	10154
2013	892 (8.1)	787 (7.2)	944 (8.6)	945 (8.6)	953 (8.7)	867 (7.9)	941 (8.6)	965 (8.8)	886 (8.1)	912 (8.3)	947 (8.6)	952 (8.7)	10991
2014	939 (8.5)	806 (7.3)	892 (8.1)	895 (8.1)	962 (8.7)	867 (7.8)	897 (8.1)	900 (8.1)	941 (8.5)	928 (8.4)	990 (8.9)	1046 (9.5)	11063
2015	929 (7.3)	982 (7.7)	1059 (8.3)	1067 (8.4)	1148 (9.0)	1035 (8.1)	1124 (8.8)	1083 (8.5)	1069 (8.4)	1052 (8.3)	1112 (8.7)	1084 (8.5)	12744
2016	1085 (8.3)	1030 (7.9)	1101 (8.4)	1070 (8.2)	1114 (8.5)	1075 (8.2)	1071 (8.2)	1076 (8.2)	1061 (8.1)	1227 (9.4)	1106 (8.4)	1094 (8.3)	13110
2017	1060 (7.4)	970 (6.8)	1077 (7.6)	1191 (8.4)	1175 (8.2)	1240 (8.7)	1259 (8.8)	1272 (8.9)	1234 (8.7)	1219 (8.6)	1227 (8.6)	1320 (9.3)	14244
2018	1328 (8.5)	1112 (7.1)	1202 (7.7)	1232 (7.9)	1317 (8.4)	1290 (8.3)	1471 (9.4)	1318 (8.4)	1217 (7.8)	1417 (9.1)	1307 (8.4)	1406 (9.0)	15617
2019	1453 (8.4)	1248 (7.2)	1380 (8.0)	1434 (8.3)	1504 (8.7)	1453 (8.4)	1463 (8.4)	1446 (8.3)	1395 (8.0)	1521 (8.8)	1543 (8.9)	1512 (8.7)	17352
2020	1573 (9.3)	1247 (7.4)	1237 (7.4)	1379 (8.2)	1443 (8.6)	1509 (9.0)	1615 (9.6)	1360 (8.1)	1277 (7.6)	1447 (8.6)	1407 (8.4)	1330 (7.9)	16824
Sum	10146	9023	9820	10003	10473	10220	10738	10220	9895	10599	10472	10490	122099

Values are presented as number (row percent)

All audited cases registered in KSR

Table 2. Number of acute ischemic stroke patients stratified by sex and age group

Age group	All (n)	Per (%)	Male(n)	Per (%)	Female(n)	Per (%)
<45	4965	4.6	3357	5.3	1608	3.6
45-54	11356	10.5	8360	13.1	2996	6.7
55-64	21940	20.2	16101	25.2	5839	13.1
65-74	29514	27.2	18031	28.2	11483	25.8
75-84	31878	29.4	15179	23.8	16699	37.4
≥85	8827	8.1	2858	4.5	5969	13.4
Mean (SD)	68.5 (13.0)		65.9 (12.5)		72.1 (12.7)	
Total	108480	100	63886	100	44594	100

Based on 108,480 ischemic stroke cases

Table 3. Secular trends of age group proportions in acute ischemic stroke

Age group	2012	2013	2014	2015	2016	2017	2018	2019	2020
Total									
<45	424 (4.9)	486 (5.1)	518 (5.3)	508 (4.6)	542 (4.8)	601 (4.8)	637 (4.5)	689 (4.3)	560 (3.7)
45-54	1003 (11.5)	1146 (11.9)	1095 (11.1)	1182 (10.8)	1207 (10.6)	1273 (10.1)	1438 (10.1)	1557 (9.8)	1455 (9.5)
55-64	1671 (19.2)	1883 (19.6)	1929 (19.6)	2171 (19.8)	2347 (20.6)	2543 (20.2)	2900 (20.4)	3294 (20.7)	3202 (20.9)
65-74	2728 (31.3)	2892 (30.1)	2840 (28.9)	3129 (28.5)	3097 (27.1)	3299 (26.2)	3597 (25.4)	4009 (25.2)	3923 (25.6)
75-84	2338 (26.8)	2605 (27.1)	2728 (27.8)	3238 (29.5)	3374 (29.6)	3850 (30.6)	4346 (30.6)	4824 (30.4)	4575 (29.9)
≥85	544 (6.2)	596 (6.2)	711 (7.2)	753 (6.9)	842 (7.4)	1014 (8.1)	1269 (8.9)	1515 (9.5)	1583 (10.3)
Men									
<45	292 (5.7)	331 (6.0)	356 (6.2)	327 (5.2)	365 (5.5)	426 (5.6)	431 (5.2)	475 (5.0)	354 (3.8)
45-54	761 (14.9)	853 (15.4)	781 (13.7)	881 (13.9)	896 (13.5)	950 (12.6)	1043 (12.6)	1135 (11.9)	1060 (11.5)
55-64	1204 (23.6)	1347 (24.3)	1379 (24.1)	1551 (24.5)	1717 (25.8)	1884 (25.0)	2137 (25.9)	2478 (26.0)	2404 (26.1)
65-74	1619 (31.7)	1664 (30.0)	1675 (29.3)	1847 (29.2)	1836 (27.6)	2058 (27.3)	2205 (26.7)	2542 (26.7)	2585 (28.1)
75-84	1041 (20.4)	1156 (20.9)	1288 (22.5)	1487 (23.5)	1585 (23.8)	1896 (25.1)	2056 (24.9)	2388 (25.1)	2282 (24.8)
≥85	183 (3.6)	193 (3.5)	241 (4.2)	238 (3.8)	260 (3.9)	331 (4.4)	387 (4.7)	495 (5.2)	530 (5.8)
Women									
<45	132 (3.7)	155 (3.8)	162 (4.0)	181 (3.9)	177 (3.7)	175 (3.5)	206 (3.5)	214 (3.4)	206 (3.4)
45-54	242 (6.7)	293 (7.2)	314 (7.7)	301 (6.5)	311 (6.5)	323 (6.4)	395 (6.7)	422 (6.6)	395 (6.5)
55-64	467 (12.9)	536 (13.2)	550 (13.4)	620 (13.3)	630 (13.3)	659 (13.1)	763 (12.9)	816 (12.8)	798 (13.1)
65-74	1109 (30.7)	1228 (30.2)	1165 (28.4)	1282 (27.6)	1261 (26.5)	1241 (24.6)	1392 (23.5)	1467 (23.0)	1338 (22.0)
75-84	1297 (35.9)	1449 (35.7)	1440 (35.1)	1751 (37.7)	1789 (37.7)	1954 (38.8)	2290 (38.6)	2436 (38.2)	2293 (37.7)
≥85	361 (10.0)	403 (9.9)	470 (11.5)	515 (11.1)	582 (12.3)	683 (13.6)	882 (14.9)	1020 (16.0)	1053 (17.3)

Based on 108,480 ischemic stroke cases

Values are presented as number (column percent)

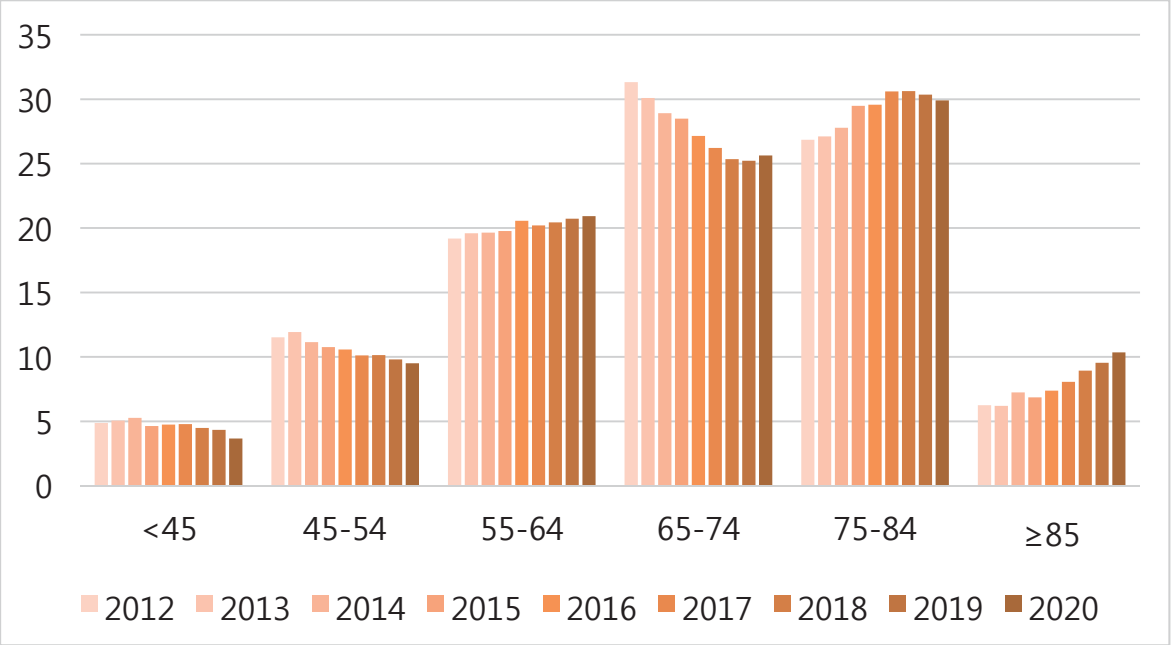


Figure 3. Secular trends of age group proportions in acute ischemic stroke

Table 4. Frequency of TOAST classification subtypes

	Frequency (n)	Percentage (%)
Large Artery Atherosclerosis	35611	32.83
Small Vessel Occlusion	21742	20.04
Cardioembolism	22148	20.42
Other Determined Etiology	4560	4.20
Undetermined Etiology - 2 or more	5570	5.13
Undetermined Etiology - Negative	10230	9.43
Undetermined Etiology - Incomplete work-ups	7659	7.06
Missing	960	0.88

Based on 122,099 ischemic stroke cases

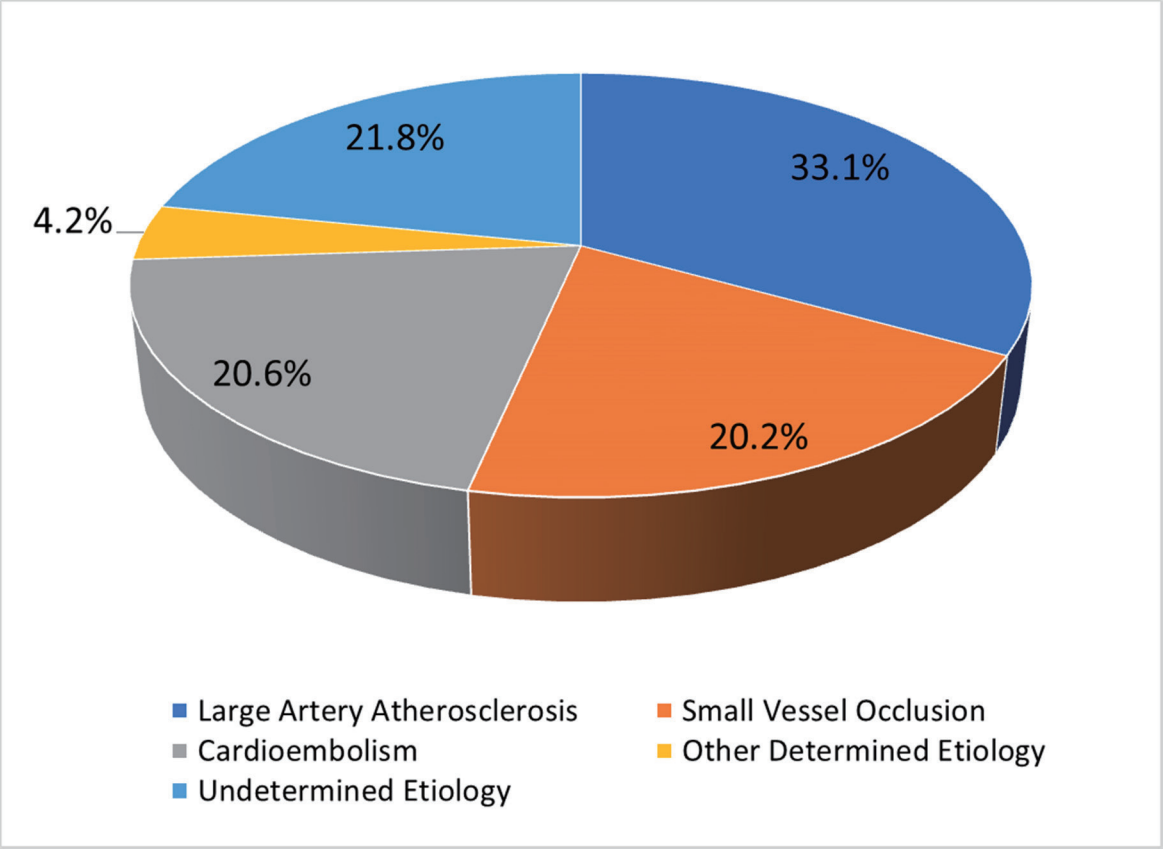


Figure 4. TOAST classification

Based on 107,520 ischemic stroke cases

Table 5. Secular trends of TOAST Classification

	Number of patients (%)								
	2012	2013	2014	2015	2016	2017	2018	2019	2020
Large Artery Atherosclerosis	3149 (37.1)	3190 (34.4)	3240 (33.4)	3522 (32.2)	3749 (33.0)	4084 (32.7)	4569 (32.3)	5004 (31.6)	5104 (33.4)
Small Vessel Occlusion	1606 (18.9)	1892 (20.4)	1821 (18.8)	2158 (19.7)	2276 (20.0)	2606 (20.9)	2953 (20.9)	3319 (20.9)	3111 (20.4)
Cardioembolism	1815 (21.4)	1976 (21.3)	2162 (22.3)	2370 (21.7)	2307 (20.3)	2467 (19.8)	2878 (20.3)	3201 (20.2)	2972 (19.5)

Other Determined Etiology	236 (2.8)	349 (3.8)	361 (3.7)	370 (3.4)	412 (3.6)	512 (4.1)	781 (5.5)	776 (4.9)	763 (5.0)
Undetermined Etiology - 2 or more	428 (5.0)	422 (4.5)	484 (5.0)	609 (5.6)	595 (5.2)	663 (5.3)	684 (4.8)	798 (5.0)	887 (5.8)
Undetermined Etiology - Negative	562 (6.6)	676 (7.3)	765 (7.9)	1052 (9.6)	1140 (10.0)	1345 (10.8)	1391 (9.8)	1675 (10.6)	1624 (10.6)
Undetermined Etiology - Incomplete work-ups	696 (8.2)	771 (8.3)	861 (8.9)	850 (7.8)	882 (7.8)	807 (6.5)	901 (6.4)	1080 (6.8)	811 (5.3)
Total	8492	9276	9694	10931	11361	12484	14157	15853	15272

Based on 107,520 ischemic stroke cases

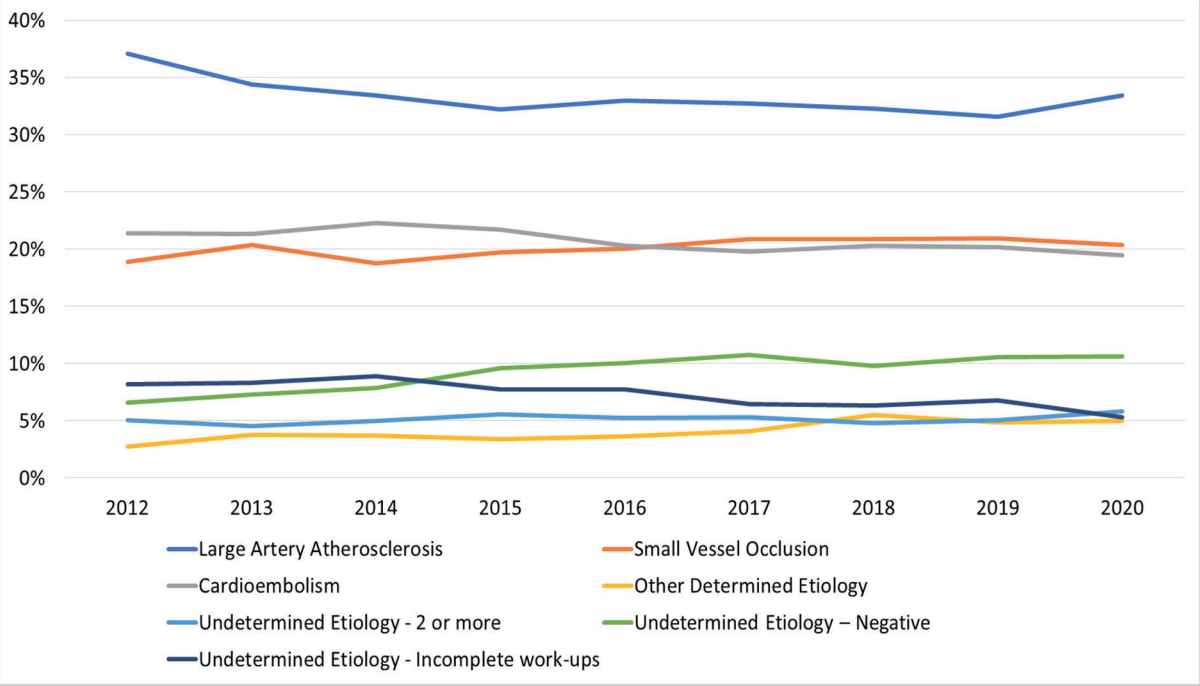


Figure 5. Secular trends of proportions in TOAST Classification

Table 6. Time interval between symptom onset and hospital arrival

	Number of prescription (%)									
	Seoul	Gyeonggi & Incheon	Gyeongnam & Busan	Chungnam & Daejeon	Gyeongbuk & Daegu	Gangwon	Gwangju & Jeonnam	Chungbuk	Jeju	Overall
≤3 hour	6895 (22.2)	7086 (24.5)	6242 (28.0)	1202 (27.6)	3787 (23.3)	721 (25.8)	2633 (27.4)	1090 (29.3)	810 (29.1)	30466 (25.0)
3-6 hour	2865 (9.2)	2920 (10.1)	2915 (13.1)	442 (10.1)	1827 (11.2)	315 (11.3)	1612 (16.8)	397 (10.7)	322 (11.6)	13615 (11.2)
6-12 hour	4121 (13.3)	3772 (13.1)	3280 (14.7)	622 (14.3)	2146 (13.2)	361 (12.9)	1483 (15.4)	531 (14.3)	404 (14.5)	16720 (13.7)
12-24 hour	5418 (17.4)	5004 (17.3)	3776 (17.0)	747 (17.1)	2980 (18.3)	500 (17.9)	1387 (14.4)	660 (17.7)	464 (16.7)	20936 (17.2)
24-36 hour	2464 (7.9)	2074 (7.2)	1459 (6.6)	350 (8.0)	1179 (7.3)	202 (7.2)	816 (8.5)	224 (6.0)	188 (6.8)	8956 (7.4)
36-48 hour	1768 (5.7)	1474 (5.1)	890 (4.0)	221 (5.1)	869 (5.3)	134 (4.8)	273 (2.8)	205 (5.5)	101 (3.6)	5935 (4.9)
2-7 day	6419 (20.6)	5431 (18.8)	3128 (14.1)	700 (16.1)	2908 (17.9)	481 (17.2)	1179 (12.3)	561 (15.1)	455 (16.4)	21262 (17.5)
>7 day	1148 (3.7)	1130 (3.9)	568 (2.6)	72 (1.7)	550 (3.4)	81 (2.9)	230 (2.4)	57 (1.5)	37 (1.3)	3873 (3.2)
TOTAL	31098	28891	22258	4356	16246	2795	9613	3725	2781	121763

Based on 121,763 acute stroke patients

Table 7. Secular trends of categorized symptom onset to arrival time

	2012	2013	2014	2015	2016	2017	2018	2019	2020
median hr (IQR)	12 (3 - 39)	12 (3 - 39)	12 (3 - 37)	12 (3 - 40)	11 (3 - 38)	12 (3 - 36)	12 (3 - 36)	12 (3 - 37)	12 (3 - 35)
≤3 hour	2404 (23.8)	2659 (24.4)	2833 (25.7)	3155 (24.8)	3362 (25.7)	3560 (25.0)	3929 (25.2)	4425 (25.5)	4139 (24.6)
3-6 hour	1231 (12.2)	1274 (11.7)	1216 (11.0)	1380 (10.9)	1462 (11.2)	1581 (11.1)	1767 (11.3)	1867 (10.8)	1837 (10.9)
6-12 hour	1364 (13.5)	1513 (13.9)	1459 (13.3)	1717 (13.5)	1839 (14.1)	1975 (13.9)	2082 (13.3)	2389 (13.8)	2382 (14.2)
12-24 hour	1696 (16.8)	1793 (16.5)	1883 (17.1)	2070 (16.3)	2179 (16.7)	2449 (17.2)	2778 (17.8)	3050 (17.6)	3038 (18.1)
24-36 hour	741 (7.3)	781 (7.2)	810 (7.4)	975 (7.7)	869 (6.7)	1088 (7.6)	1174 (7.5)	1234 (7.1)	1284 (7.6)
36-48 hour	507 (5.0)	509 (4.7)	535 (4.9)	690 (5.4)	625 (4.8)	717 (5.0)	750 (4.8)	825 (4.8)	777 (4.6)
2-7 day	1839 (18.2)	1998 (18.3)	1894 (17.2)	2271 (17.9)	2273 (17.4)	2394 (16.8)	2690 (17.2)	3061 (17.6)	2842 (16.9)
>7 day	317 (3.1)	362 (3.3)	380 (3.5)	441 (3.5)	457 (3.5)	462 (3.2)	444 (2.8)	493 (2.8)	517 (3.1)

Based on 121,763 acute stroke patients

Table 7-1. Secular trends of categorized symptom onset (last normal time) to arrival time

	2012	2013	2014	2015	2016	2017	2018	2019	2020
median hr (IQR)	12 (3 - 39)	12 (3 - 39)	12 (3 - 37)	12 (3 - 40)	11 (3 - 38)	12 (3 - 36)	12 (3 - 36)	12 (3 - 37)	12 (3 - 35)
≤2 hour	1787 (17.7)	2007 (18.4)	2104 (19.1)	2388 (18.8)	2610 (20.0)	2667 (18.7)	2995 (19.2)	3436 (19.8)	3180 (18.9)
≤3.5 hour	2670 (26.4)	2919 (26.8)	3065 (27.8)	3480 (27.4)	3675 (28.1)	3865 (27.2)	4323 (27.7)	4817 (27.8)	4541 (27.0)
≤24 hour	6695 (66.3)	7239 (66.5)	7391 (67.1)	8320 (65.5)	8842 (67.7)	9565 (67.2)	10556 (67.6)	11731 (67.6)	11396 (67.8)

Based on 121,758 acute stroke patients

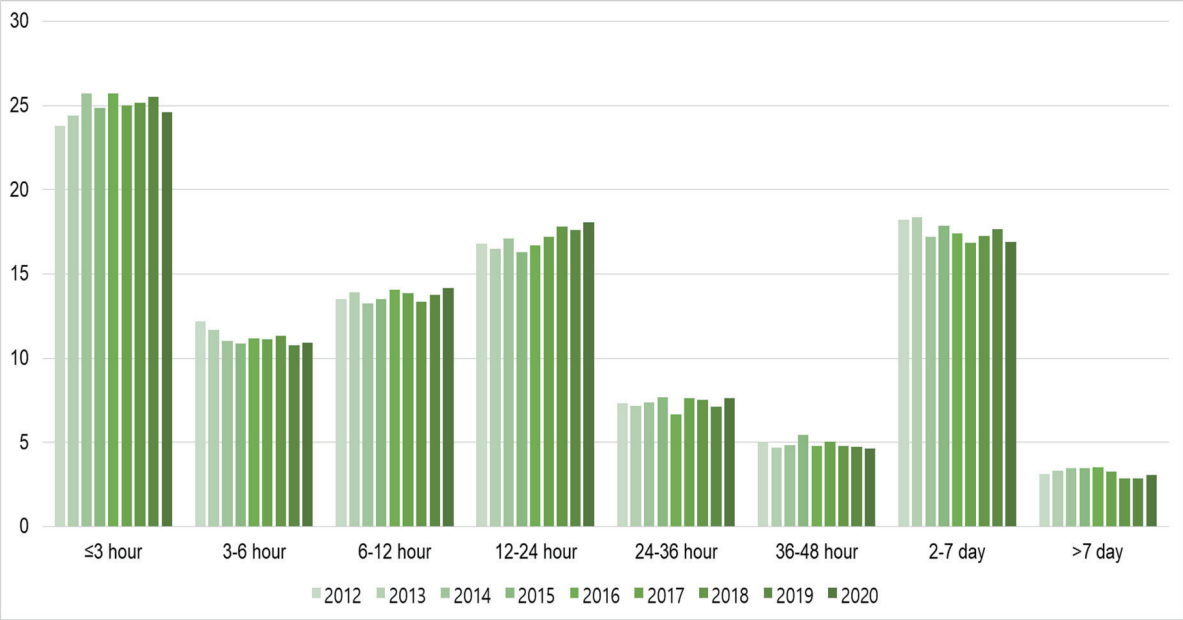


Figure 6. Secular trends of categorized symptom onset to arrival time

Table 8. Secular trends of anthropometric measurements of acute ischemic stroke and transient ischemic attack patients

	Overall, mean (SD)				
	2012-2013	2014-2015	2016-2017	2018-2019	2020
Height (cm)	162.2 (9.0)	162.2 (9.0)	162.5 (9.0)	162.6 (9.0)	162.7 (9.0)
Weight (kg)	62.5 (11.5)	62.6 (11.8)	63.2 (12.1)	63.3 (12.2)	63.6 (12.1)
BMI (kg/m^2)	23.7 (3.4)	23.7 (3.5)	23.8 (3.5)	23.8 (3.6)	23.9 (3.6)
Initial SBP (mmHg)	146.5 (27.1)	146.2 (27.1)	148.8 (27.7)	151.3 (28.4)	153.5 (28.9)
Initial DBP (mmHg)	84.8 (15.2)	84.5 (15.4)	84.6 (15.9)	85.1 (16.5)	86.2 (16.8)
	Men, mean (SD)				
Height (cm)	167.9 (6.1)	167.7 (6.4)	168.0 (6.3)	168.0 (6.3)	168.0 (6.3)
Weight (kg)	67.1 (10.4)	67.3 (10.6)	67.9 (10.9)	68.0 (11.1)	68.2 (11.0)
BMI (kg/m^2)	23.8 (3.2)	23.9 (3.2)	24.0 (3.2)	24.0 (3.3)	24.1 (3.3)
Initial SBP (mmHg)	145.9 (26.4)	146.2 (26.5)	148.7 (27.1)	150.9 (27.8)	153.0 (28.3)
Initial DBP (mmHg)	85.2 (14.9)	85.0 (15.2)	85.3 (15.8)	85.8 (16.5)	86.8 (16.8)
	Women, mean (SD)				
Height (cm)	154.4 (6.0)	154.7 (6.2)	154.7 (6.1)	154.8 (6.1)	154.7 (6.2)
Weight (kg)	56.0 (9.8)	56.2 (10.1)	56.5 (10.4)	56.4 (10.3)	56.7 (10.3)
BMI (kg/m^2)	23.5 (3.7)	23.4 (3.8)	23.5 (3.8)	23.5 (3.9)	23.7 (3.9)
Initial SBP (mmHg)	147.2 (27.9)	146.3 (28.0)	148.9 (28.6)	151.9 (29.2)	154.1 (29.8)
Initial DBP (mmHg)	84.2 (15.5)	83.9 (15.6)	83.5 (16.0)	84.0 (16.4)	85.2 (16.7)

Based on 116,590 acute ischemic stroke and transient ischemic attack patients
Note: The outlier cases are excluded from the analysis in the following cases: height, not between 100-250; weight, not between 20-200; initial systolic BP, not between 50-300; initial diastolic BP, more than 250.
BMI: body mass index; SBP: systolic blood pressure; DBP: diastolic blood pressure.

Table 9. Secular trends of prestroke mRS score

Prestroke mRS score	Number (%)					
	Total	2012-2013	2014-2015	2016-2017	2018-2019	2020
0	83590 (77.1)	14234 (77.7)	14712 (70.7)	19163 (79.9)	23666 (78.7)	11815 (77.2)
1	9279 (8.6)	1332 (7.3)	2083 (10.0)	1768 (7.4)	2614 (8.7)	1482 (9.7)
2	6082 (5.6)	911 (5.0)	1709 (8.2)	1258 (5.2)	1396 (4.6)	808 (5.3)
3	5373 (5.0)	954 (5.2)	1249 (6.0)	1046 (4.4)	1425 (4.7)	699 (4.6)
4	2812 (2.6)	564 (3.1)	660 (3.2)	549 (2.3)	692 (2.3)	347 (2.3)
5	1341 (1.2)	319 (1.7)	388 (1.9)	205 (0.9)	282 (0.9)	147 (1.0)

Based on 108,477 ischemic stroke cases

Table 10. Secular trends of history of major vascular risk factors

Risk factors and prior managements	Overall	2012	2013	2014	2015	2016	2017	2018	2019	2020
Number of patients	108480	8708	9608	9821	10981	11409	12580	14187	15888	15298
Hypertension	71724 (66.1)	5927 (68.1)	6385 (66.5)	6507 (66.3)	6911 (62.9)	7389 (64.8)	8170 (64.9)	9438 (66.5)	10614 (66.8)	10383 (67.9)
History of Hypertension	64463 (91.1)	5270 (88.9)	5789 (90.7)	6005 (92.3)	6445 (93.3)	6806 (92.1)	7324 (91.8)	8151 (90.6)	9489 (90.9)	9184 (89.5)
Diagnosed at the index admission	6333 (8.9)	656 (11.1)	596 (9.3)	502 (7.7)	466 (6.7)	583 (7.9)	655 (8.2)	844 (9.4)	950 (9.1)	1081 (10.5)
Diabetes	35623 (32.8)	2910 (33.4)	3183 (33.1)	3196 (32.5)	3437 (31.3)	3652 (32.0)	4070 (32.4)	4671 (32.9)	5263 (33.1)	5241 (34.3)
History of Diabetes	30687 (87.2)	2529 (86.9)	2726 (85.6)	2843 (89.0)	3125 (90.9)	3250 (89.0)	3468 (87.1)	3825 (85.9)	4516 (87.0)	4405 (85.0)
Diagnosed at the index admission	4494 (12.8)	380 (13.1)	457 (14.4)	353 (11.0)	312 (9.1)	401 (11.0)	514 (12.9)	627 (14.1)	675 (13.0)	775 (15.0)
Dyslipidemia	35774 (33.0)	2787 (32.0)	2770 (28.8)	2818 (28.7)	2945 (26.8)	3493 (30.6)	3791 (30.1)	4811 (33.9)	6176 (38.9)	6183 (40.4)
History of Dyslipidemia	22201 (63.4)	1462 (52.5)	1648 (59.5)	1748 (62.0)	1968 (66.8)	2399 (68.7)	2516 (68.7)	2869 (64.8)	3742 (62.3)	3849 (63.1)
Diagnosed at the index admission	12804 (36.6)	1324 (47.5)	1122 (40.5)	1070 (38.0)	977 (33.2)	1094 (31.3)	1145 (31.3)	1558 (35.2)	2267 (37.7)	2247 (36.9)
Current smoker	24966 (23.1)	2221 (25.7)	2425 (25.4)	2339 (23.9)	2267 (21.0)	2521 (22.4)	2867 (22.8)	3106 (21.9)	3646 (23.0)	3574 (23.4)
Atrial fibrillation	21985 (20.3)	1811 (20.8)	1887 (19.6)	2089 (21.3)	2318 (21.1)	2269 (19.9)	2411 (19.2)	2899 (20.4)	3199 (20.1)	3102 (20.3)
History of Atrial fibrillation	11907 (54.8)	1074 (59.3)	1074 (56.9)	1132 (54.2)	1397 (60.3)	1319 (58.4)	1277 (54.4)	1491 (53.5)	1614 (51.1)	1529 (49.8)
Diagnosed at the index admission	9819 (45.2)	737 (40.7)	812 (43.1)	957 (45.8)	919 (39.7)	940 (41.6)	1072 (45.6)	1298 (46.5)	1542 (48.9)	1542 (50.2)

Based on 108,480 ischemic stroke cases. Percentages of management categories were calculated based on the frequency of each risk factor.

Table 10-1. Regional variation of vascular risk factors and their management before index stroke

Risk factors and prior managements	Seoul	Gyeonggi & Incheon	Gyeongnam & Busan	Chungnam & Daejeon	Gyeongbuk & Daegu	Gangwon	Gwangju & Jeonnam	Chungbuk	Jeju
Number of patients	28541	25880	18758	3888	13978	2524	9323	2991	2597
Hypertension	19599 (68.7)	17783 (68.7)	12104 (64.5)	2637 (67.8)	9188 (65.7)	1571 (62.2)	5312 (57.0)	1843 (61.6)	1687 (65.0)
History of Hypertension	17126 (91.3)	16015 (90.1)	10466 (86.7)	2470 (93.7)	8312 (90.9)	1510 (96.1)	5226 (98.4)	1769 (96.0)	1569 (93.0)
Diagnosed at the index admission	1626 (8.7)	1766 (9.9)	1608 (13.3)	166 (6.3)	828 (9.1)	61 (3.9)	86 (1.6)	74 (4.0)	118 (7.0)
Diabetes	10040 (35.2)	8810 (34.0)	5949 (31.7)	1209 (31.1)	4431 (31.7)	854 (33.8)	2597 (27.9)	963 (32.2)	770 (29.6)
History of Diabetes	8446 (87.6)	7417 (84.2)	4994 (84.2)	1072 (88.9)	3934 (89.3)	768 (89.9)	2509 (96.6)	918 (95.3)	629 (81.7)
Diagnosed at the index admission	1200 (12.4)	1393 (15.8)	937 (15.8)	134 (11.1)	470 (10.7)	86 (10.1)	88 (3.4)	45 (4.7)	141 (18.3)
Dyslipidemia	12004 (42.1)	9632 (37.2)	5785 (30.8)	712 (18.3)	4739 (33.9)	408 (16.2)	1348 (14.5)	386 (12.9)	760 (29.3)
History of Dyslipidemia	6859 (60.7)	6919 (71.8)	2781 (48.1)	667 (93.8)	2917 (62.2)	362 (88.7)	858 (63.6)	304 (78.8)	534 (70.3)
Diagnosed at the index admission	4434 (39.3)	2713 (28.2)	2997 (51.9)	44 (6.2)	1772 (37.8)	46 (11.3)	490 (36.4)	82 (21.2)	226 (29.7)
Current smoker	6607 (23.2)	6503 (25.2)	4809 (25.8)	854 (22.0)	2836 (20.7)	359 (14.7)	1762 (18.9)	611 (20.4)	625 (24.1)
Atrial fibrillation	5374 (18.8)	5282 (20.4)	4006 (21.4)	767 (19.7)	2566 (18.4)	477 (18.9)	2290 (24.6)	639 (21.4)	584 (22.5)
History of Atrial fibrillation	2901 (56.2)	2947 (55.8)	1781 (44.5)	371 (48.4)	1479 (58.7)	265 (55.6)	1450 (63.3)	394 (61.7)	319 (54.6)
Diagnosed at the index admission	2265 (43.8)	2335 (44.2)	2220 (55.5)	395 (51.6)	1042 (41.3)	212 (44.4)	840 (36.7)	245 (38.3)	265 (45.4)

Table 11. NIHSS score at hospital arrival

Region	Mean (SD)	Median (IQR)	Min-Max
Seoul	5.2 (5.6)	3 (1 - 7)	0 - 40
Gyeonggi & & Incheon	5.6 (6.2)	3 (1 - 7)	0 - 42
Gyeongnam & Busan	5.9 (6.1)	4 (2 - 8)	0 - 38
Chungnam & Daejeon	5.4 (6.1)	3 (1 - 7)	0 - 38
Gyeongbuk & Daegu	5.3 (6.1)	3 (1 - 7)	0 - 42
Gangwon	4.9 (5.6)	3 (1 - 6)	0 - 30
Gwangju & Jeonnam	5.3 (5.6)	3 (1 - 8)	0 - 31
Chungbuk	5.5 (5.7)	3 (2 - 7)	0 - 35
Jeju	6.8 (6.9)	4 (2 - 10)	0 - 38
Total	5.5 (6.0)	3 (1 - 7)	0 - 42

Based on 108,479 ischemic stroke cases. NIHSS: National Institutes of Health Stroke Scale

Table 11-1. Secular trends of proportion in minor (NIHSS≤3) and severe stroke (NIHSS>15)

Year of admission	Overall	2012	2013	2014	2015	2016	2017	2018	2019	2020
NIHSS, median (IQR)	3 (1 - 7)	4 (1 - 8)	3 (1 - 7)	4 (1 - 8)	3 (1 - 7)	3 (1 - 8)	3 (1 - 7)	3 (1 - 7)	3 (1 - 7)	3 (1 - 7)
NIHSS <4, %	55587 (51.2)	4306 (49.4)	4883 (50.8)	4823 (49.1)	5738 (52.3)	5842 (51.2)	6449 (51.3)	7174 (50.6)	8408 (52.9)	7964 (52.1)
NIHSS ≥15, %	11764 (10.8)	973 (11.2)	1094 (11.4)	1210 (12.3)	1232 (11.2)	1246 (10.9)	1307 (10.4)	1529 (10.8)	1628 (10.2)	1545 (10.1)

Table 12. Profile of method of revascularization therapies for ischemic strokes

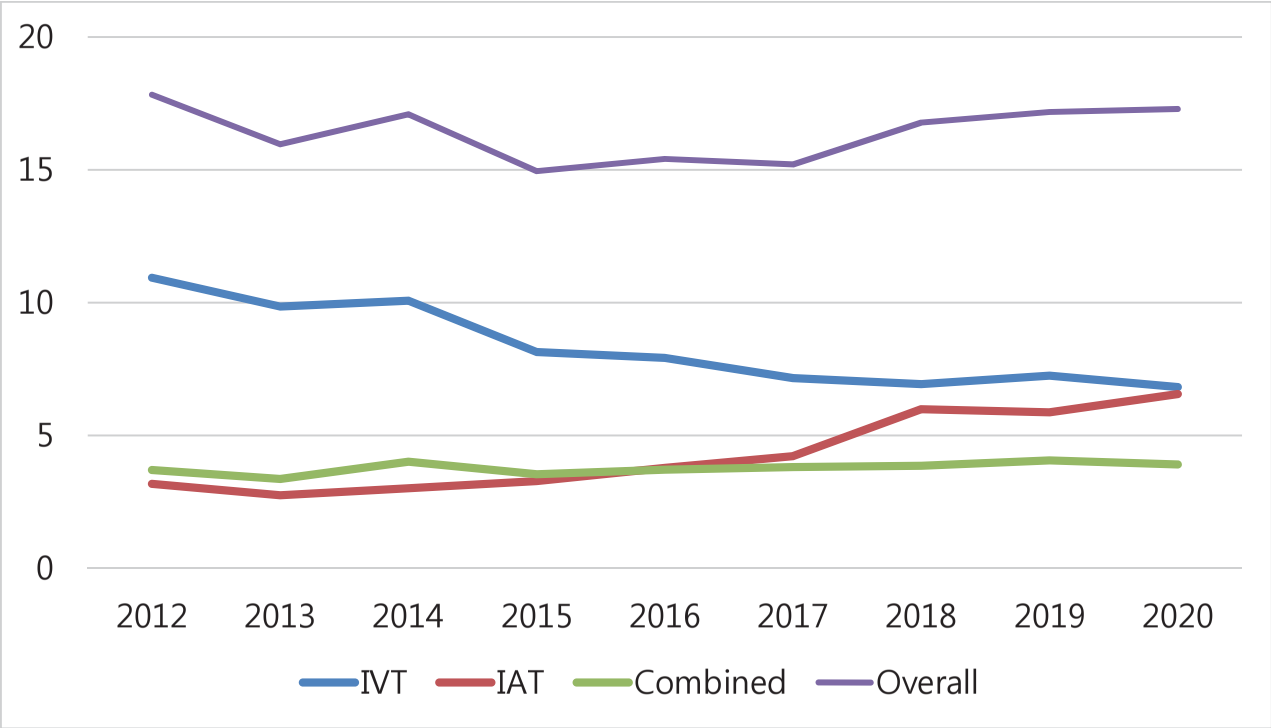
Region	IVT	IAT	Combined	Total
Seoul	1830 (6.8)	804 (3.0)	710 (2.6)	3344 (12.3)
Gyeonggi & & Incheon	1769 (7.0)	1247 (4.9)	986 (3.9)	4002 (15.8)
Gyeongnam & Busan	1888 (10.9)	1063 (6.2)	918 (5.3)	3869 (22.4)
Chungnam & Daejeon	358 (9.2)	67 (1.7)	110 (2.8)	535 (13.8)
Gyeongbuk & Daegu	833 (6.1)	781 (5.8)	463 (3.4)	2077 (15.3)
Gangwon	211 (10.1)	56 (2.7)	79 (3.8)	346 (16.6)
Gwangju & Jeonnam	959 (10.3)	477 (5.1)	389 (4.2)	1825 (19.6)
Chungbuk	187 (6.3)	173 (5.8)	166 (5.6)	526 (17.7)
Jeju	286 (11.0)	144 (5.5)	142 (5.5)	572 (22.0)
Total	8321 (8.0)	4812 (4.6)	3963 (3.8)	17096 (16.4)

Based on 104,127 ischemic stroke cases. IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Table 13. Secular trends of revascularization modalities

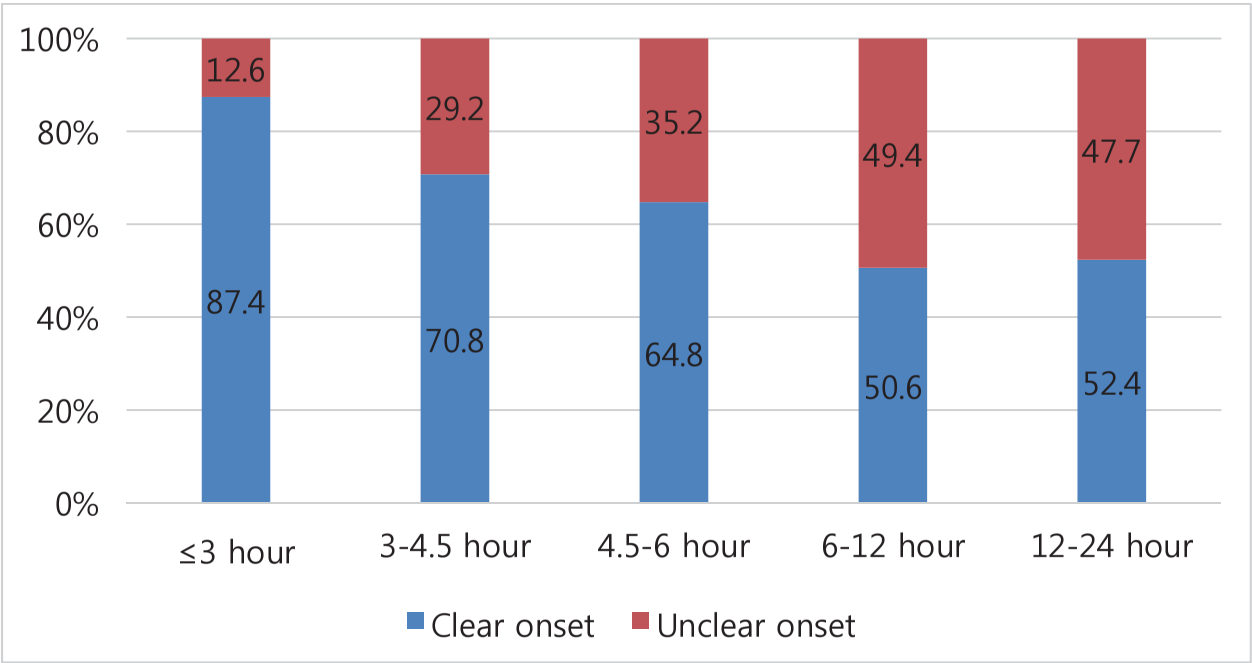
	2012	2013	2014	2015	2016	2017	2018	2019	2020	Total
IVT	730 (10.9)	815 (9.8)	932 (10.1)	876 (8.1)	894 (7.9)	896 (7.2)	983 (6.9)	1152 (7.3)	1043 (6.8)	8321 (8.0)
IAT	212 (3.2)	228 (2.8)	279 (3.0)	353 (3.3)	427 (3.8)	528 (4.2)	850 (6.0)	932 (5.9)	1003 (6.6)	4812 (4.6)
Combined	247 (3.7)	278 (3.4)	371 (4.0)	381 (3.5)	419 (3.7)	477 (3.8)	547 (3.9)	645 (4.1)	598 (3.9)	3963 (3.8)
Overall	1189 (17.8)	1321 (16.0)	1582 (17.1)	1610 (15.0)	1740 (15.4)	1901 (15.2)	2380 (16.8)	2729 (17.2)	2644 (17.3)	17096 (16.4)

Based on 104,127 ischemic stroke cases. IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis



Based on 104,127 ischemic stroke cases. IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Figure 7. Secular trends of revascularization modalities



Based on 71,137 ischemic stroke cases

Figure 8. Proportions of clear onset, unclear onset by categorized onset (LNT) to arrival time

Table 14. Secular trends of laboratory tests

	Mean (SD)					
	Overall	2012-2013	2014-2015	2016-2017	2018-2019	2020
Blood Urea Nitrogen (mg/dL)	17.6 (9.2)	17.5 (9.0)	17.5 (9.3)	17.5 (9.3)	17.6 (9.0)	18.0 (9.6)
Creatinine (mg/dL)	1.01 (1.29)	1.01 (0.92)	1.01 (1.42)	1.02 (1.11)	0.99 (0.96)	1.02 (2.06)
Hemoglobin (g/dL)	13.6 (2.1)	13.6 (2.0)	13.6 (2.0)	13.6 (2.1)	13.6 (2.1)	13.6 (2.1)
TG	126.3 (90.6)	122.8 (85.7)	122.1 (82.9)	125.8 (85.4)	129.5 (96.3)	130.3 (100.8)
LDL-cholesterol (mg/dL)	106.6 (38.2)	109.3 (37.8)	107.8 (37.9)	105.8 (38.1)	105.3 (37.4)	105.6 (40.8)
HDL-cholesterol (mg/dL)	45.6 (13.6)	44.7 (13.9)	45.1 (13.0)	45.4 (14.5)	46.0 (13.3)	47.0 (13.3)
Platelet count ($x10^3\mu L$)	230.5 (79.0)	231.4 (82.5)	226.4 (87.5)	231.1 (77.2)	232.3 (74.8)	229.9 (73.8)
Total cholesterol (mg/dL)	172.4 (45.1)	175.2 (43.6)	173.8 (44.0)	171.7 (45.5)	170.4 (45.3)	172.2 (47.4)
Initial Glucose (mg/dL)	145.6 (64.5)	144.6 (63.7)	144.9 (65.9)	145.2 (63.5)	146.2 (64.6)	146.9 (64.7)
Fasting Glucose (mg/dL)	123.7 (50.8)	125.3 (53.3)	121.1 (50.0)	121.7 (48.5)	124.7 (50.6)	126.4 (52.3)
Hb A1c (%)	6.34 (1.95)	6.44 (1.43)	6.36 (1.61)	6.31 (1.87)	6.31 (2.54)	6.34 (1.47)

Based on 108,480 ischemic stroke cases

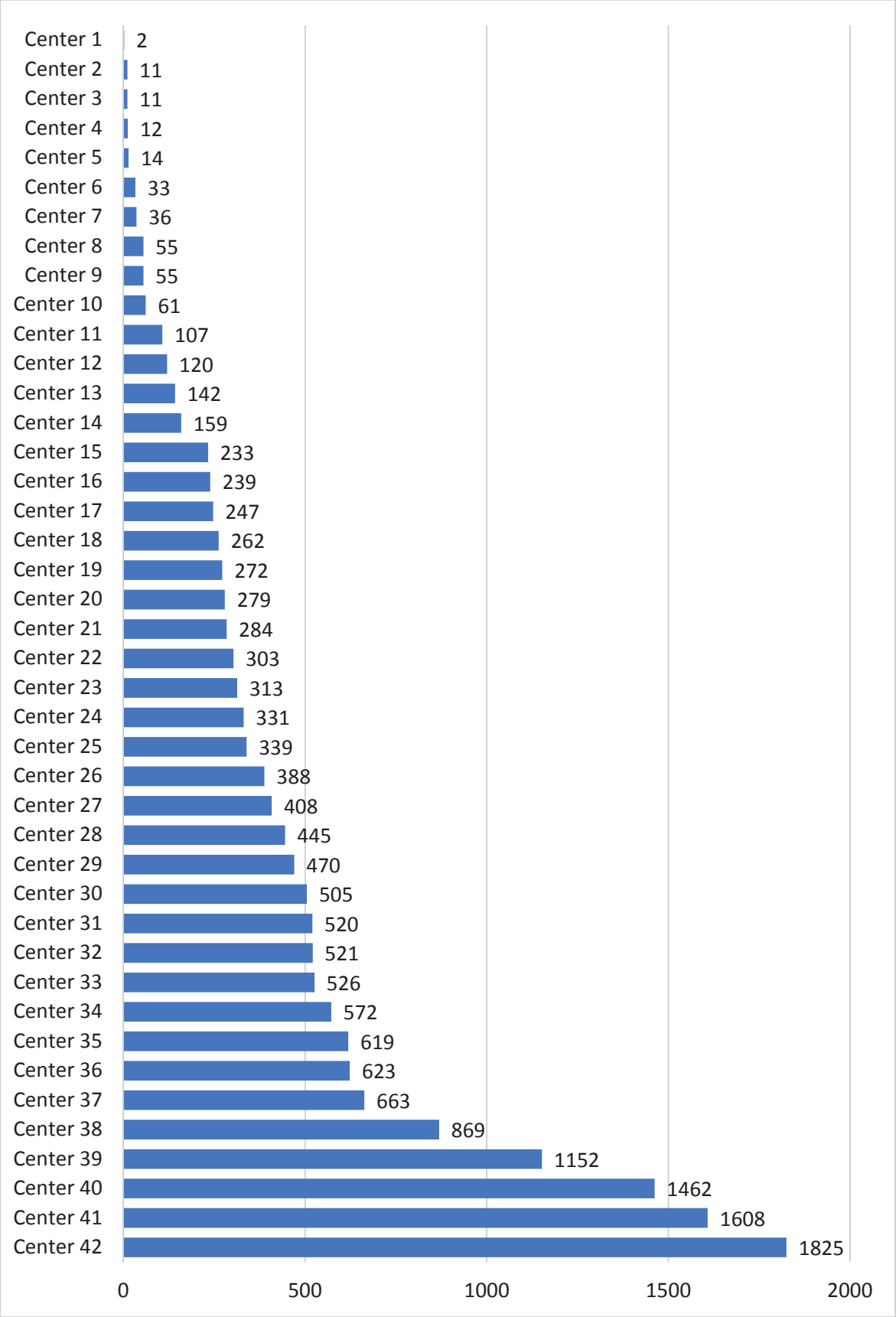


Figure 9. Volume of acute revascularization treatment for each stroke center (Anonymized)

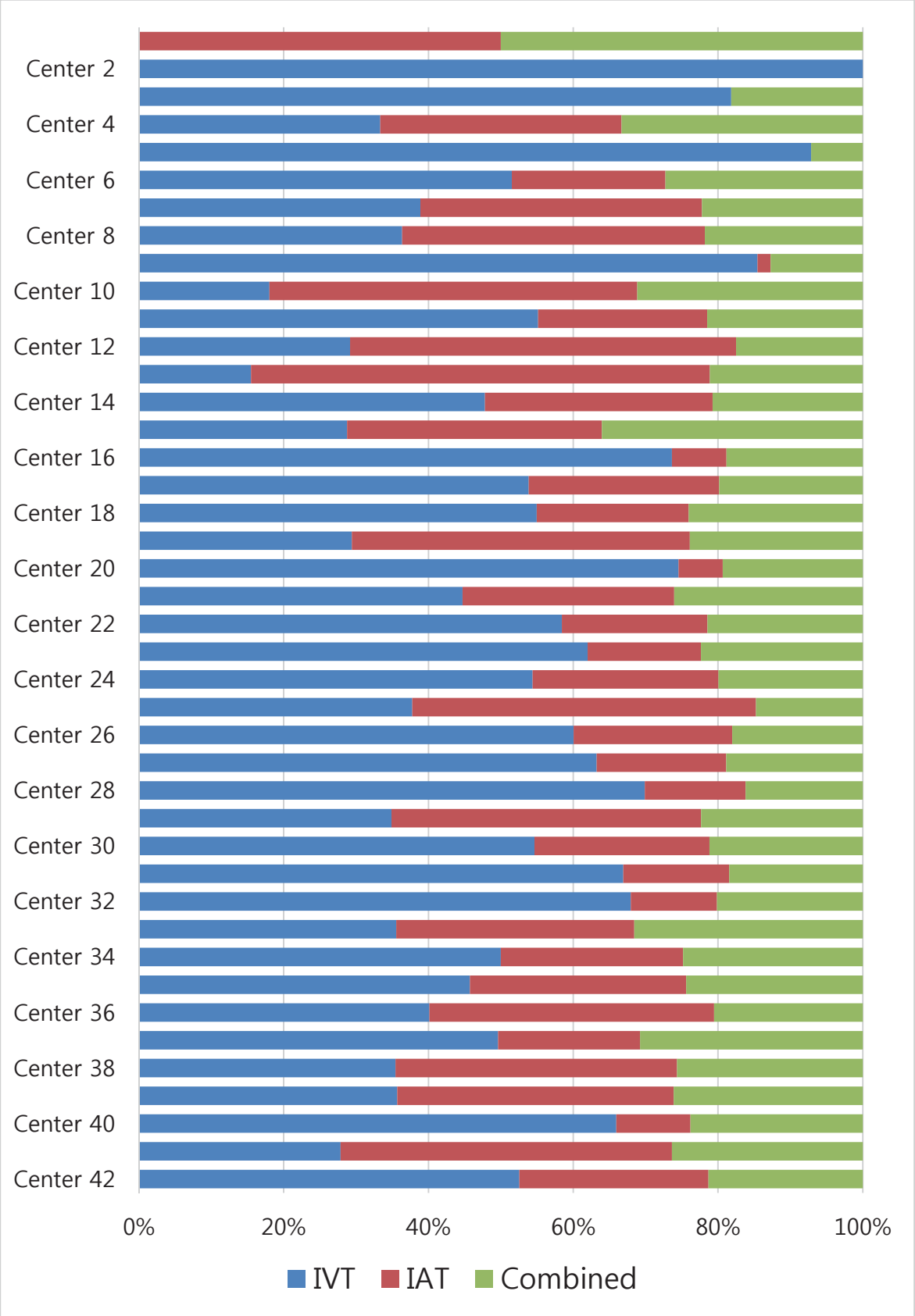


Figure 10. Discrepancy in revascularization modalities among centers

IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Table 15. Regional disparity of time interval between symptom onset and hospital arrival

	Percent of patients, %								
	Seoul	Gyeonggi & Incheon	Gyeongnam & Busan	Chungnam & Daejeon	Gyeongbuk & Daegu	Gangwon	Gwangju & Jeonnam	Chungbuk	Jeju
≤3 hour	20.7	22.4	25.6	25.2	20.6	25.2	26.7	25.3	26.5
3-6 hour	8.9	9.8	12.8	9.9	10.9	11.1	16.8	9.9	11.6
6-12 hour	13.4	13.3	15.0	14.4	13.5	13.0	15.5	14.8	15.1
12-24 hour	18.0	18.1	17.8	18.1	19.4	18.2	14.6	19.1	17.5
24-36 hour	8.2	7.5	6.9	8.4	7.6	7.3	8.7	6.6	7.1
36-48 hour	5.9	5.4	4.3	5.4	5.6	4.8	2.9	6.2	3.8
2-7 day	21.2	19.6	14.9	17.0	18.9	17.5	12.5	16.6	17.0
>7 day	3.7	3.9	2.8	1.7	3.5	3.0	2.4	1.5	1.4

Based on 108,225 ischemic stroke cases

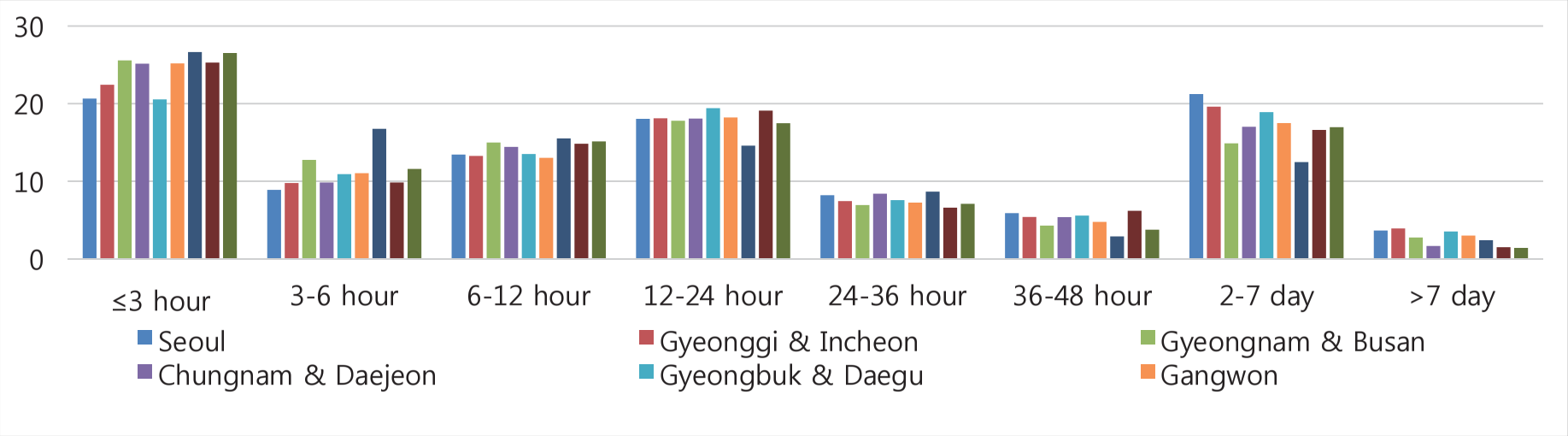


Figure 11. Regional disparity of time interval between symptom onset and hospital arrival

Table 16. Distribution and secular trends of mRS score at discharge

mRS	Number of patients (%)									
	Overall	2012	2013	2014	2015	2016	2017	2018	2019	2020
0	16059 (15.1)	1261 (15.3)	1295 (14.1)	1515 (16.2)	1730 (16.1)	1993 (17.7)	1893 (15.3)	2067 (14.6)	2280 (14.4)	2025 (13.3)
1	27504 (25.8)	2054 (25.0)	2211 (24.0)	2292 (24.6)	2669 (24.9)	2760 (24.5)	3162 (25.6)	3632 (25.6)	4545 (28.6)	4179 (27.4)
2	20818 (19.6)	1554 (18.9)	1991 (21.6)	1862 (20.0)	2309 (21.6)	2215 (19.7)	2478 (20.1)	2652 (18.7)	2848 (17.9)	2909 (19.0)
3	16178 (15.2)	1292 (15.7)	1482 (16.1)	1372 (14.7)	1609 (15.0)	1625 (14.4)	1814 (14.7)	2306 (16.3)	2377 (15.0)	2301 (15.1)
4	14435 (13.6)	1189 (14.5)	1284 (13.9)	1332 (14.3)	1410 (13.2)	1527 (13.6)	1629 (13.2)	1818 (12.8)	2145 (13.5)	2101 (13.8)
5	9481 (8.9)	813 (9.9)	853 (9.3)	849 (9.1)	837 (7.8)	932 (8.3)	1127 (9.1)	1351 (9.5)	1345 (8.5)	1374 (9.0)
6	1965 (1.8)	59 (0.7)	101 (1.1)	110 (1.2)	149 (1.4)	216 (1.9)	247 (2.0)	351 (2.5)	344 (2.2)	388 (2.5)

Based on 106,440 ischemic stroke cases. mRS: modified Rankin Scale

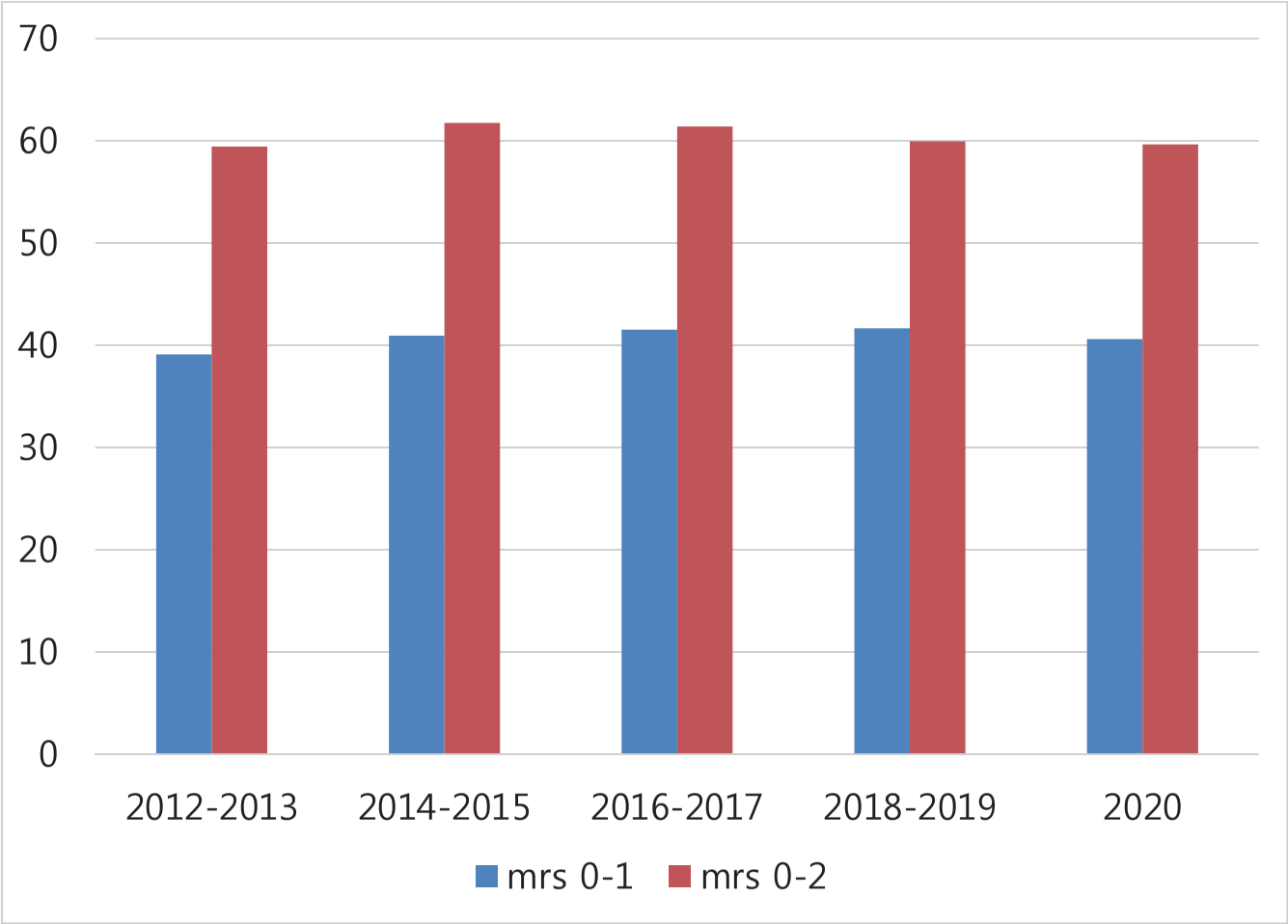


Figure 12. secular trends of proportion in mRS 0-1 and 0-2 score at discharge